Hardware Setup and Interconnection

TinyML-Based Coffee Soil Suitability Prediction System

The hardware architecture separates soil-data acquisition and machine-learning processing from TFT display control. An ESP32 handles the NPK sensor interface, rotary encoder and inference functions, while an ATmega328P drives the parallel TFT interface. The controllers communicate through a bidirectional UART link with 3.3 V/5 V logic-level translation.

1. ESP32, Rotary Encoder and Sensor Interface

The ESP32 operates at 3.3 V and provides the main data-processing interface. The five-pin rotary encoder header supplies 3.3 V and ground and carries CLK, DT and SW signals to ESP32 GPIO inputs for menu navigation. A separate five-pin sensor connector provides power, ground, NPK_RX, NPK_TX and a combined RE/DE control signal for the intended RS485 communication interface. The ESP32 also exchanges serial data with the ATmega328P through the TX_3.3V and RX_3.3V nets. A TXS0108E level-shifter stage connects these to the corresponding 5 V UART signals.

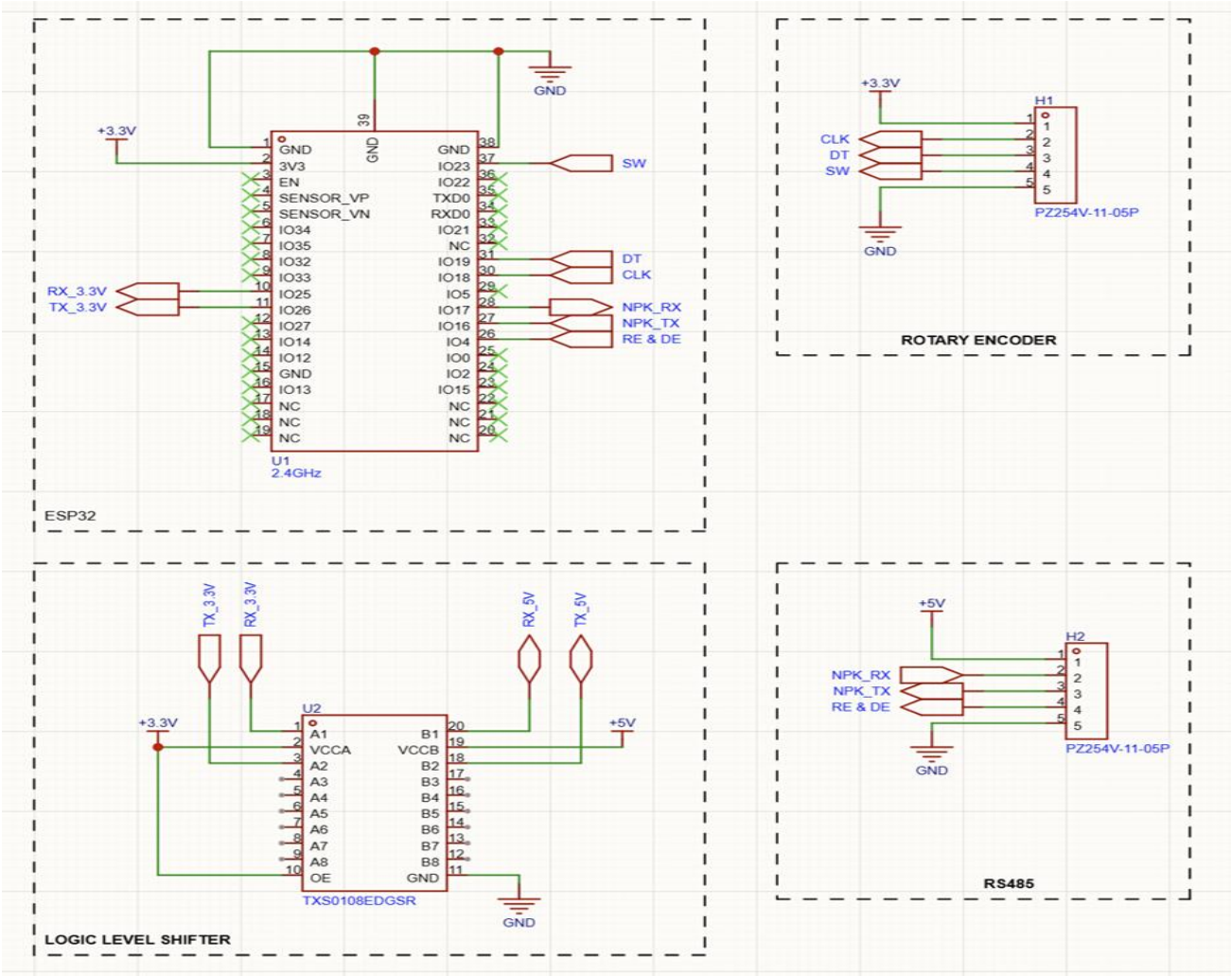


Figure 1. ESP32, rotary encoder, sensor connector and UART logic-level interface.

2. ATmega328P and TFT Display Interface

The ATmega328P is supplied from the regulated 5 V rail and uses an external 16 MHz crystal with two 22 pF load capacitors. Its hardware UART pins (PD0/RXD and PD1/TXD) connect to the translated 5 V serial nets, allowing it to receive display information from the ESP32. The microcontroller drives the TFT through parallel LCD data and control lines routed to the display connectors, including LCD_D0–LCD_D7, LCD_CS, LCD_RS, LCD_WR, LCD_RD and LCD_RST. This arrangement delegates TFT refresh and user-interface rendering to the ATmega328P.

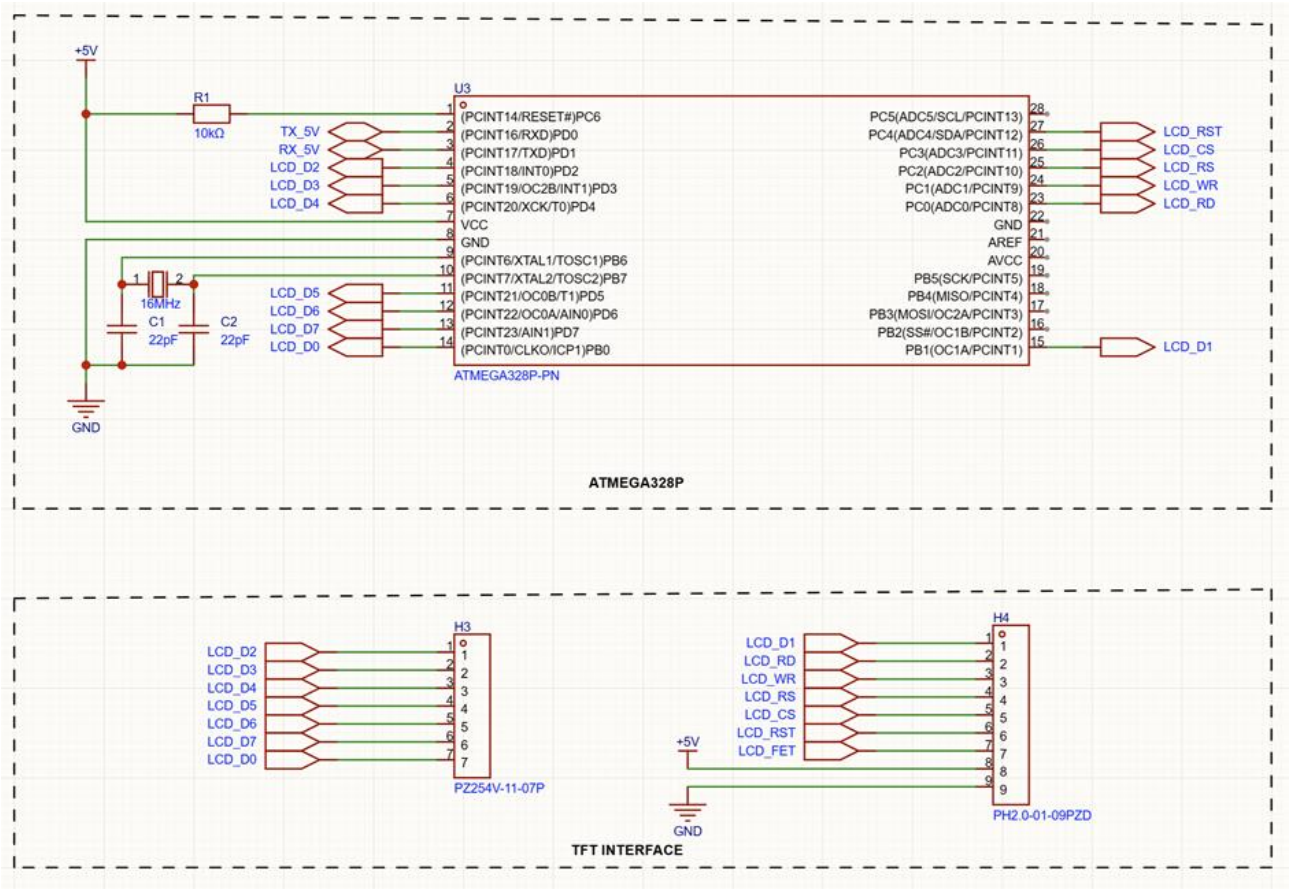


Figure 2. ATmega328P controller, crystal circuit and parallel TFT interface.

3. Power Supply and Voltage Regulation

The power section accepts an external DC input through the barrel connector. An LM2596S-5.0 switching regulator provides the 5 V rail for the ATmega328P, display-side electronics and other 5 V circuits. An RT9080-33 linear regulator derives a 3.3 V rail for the ESP32 and low-voltage interfaces. The schematic includes input/output filtering capacitors, protection components and test points for verifying both regulated supplies. A common ground reference is shared across the controllers, level shifter and sensor interface.

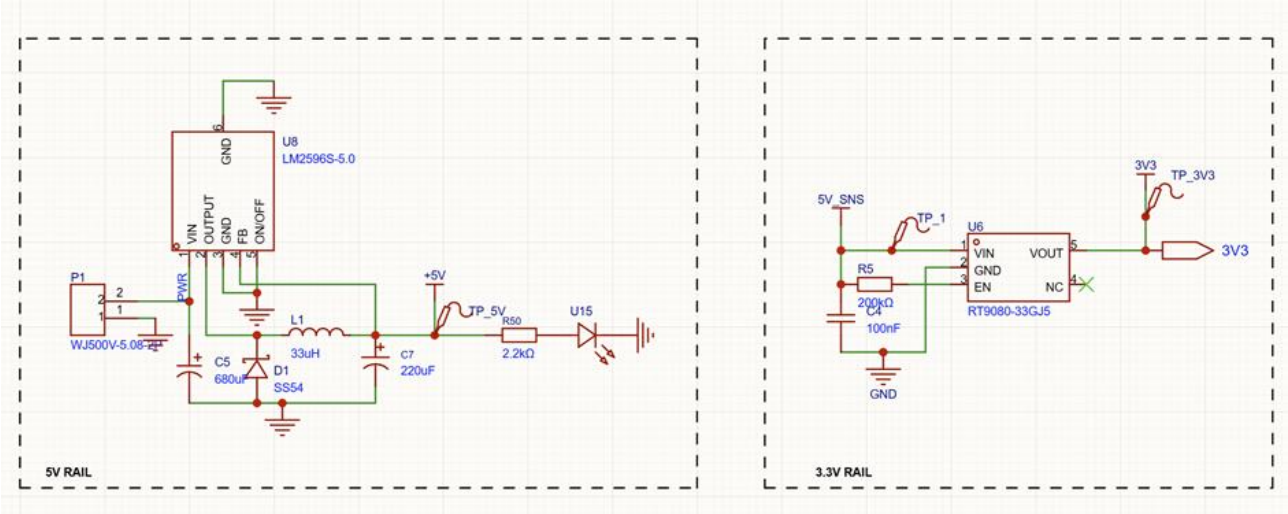


Figure 3. DC input, regulated 5 V rail and 3.3 V supply.

4. Integration Notes

Before PCB fabrication, verify that the NPK sensor connection includes a suitable RS485 transceiver: the connector shown carries UART-style RX/TX and RE/DE signals, not the differential A/B pair itself. Also confirm the TFT module voltage requirements, logic-level-shifter suitability for push-pull UART, power-regulator component values and available current, and the ATmega328P reset/decoupling arrangement. These checks are important for reliable operation.

5. Prototype Hardware Implementation

The photographs below document the physical prototype corresponding to the circuit schematics. They complement the electrical drawings by showing the assembled controller, sensor connections, user input and TFT interface.

5.1 Integrated System and Soil Measurement Setup



Figure 4. Assembled coffee soil suitability prototype during soil-sensor testing, showing the soil probe, controller board, TFT display and rotary encoder.

The NPK probe is inserted into a soil sample, while the main electronics process the sensor readings and present system information on the TFT display. The rotary encoder provides a local user-input control.

5.2 TFT Controller and Prototype Interconnections

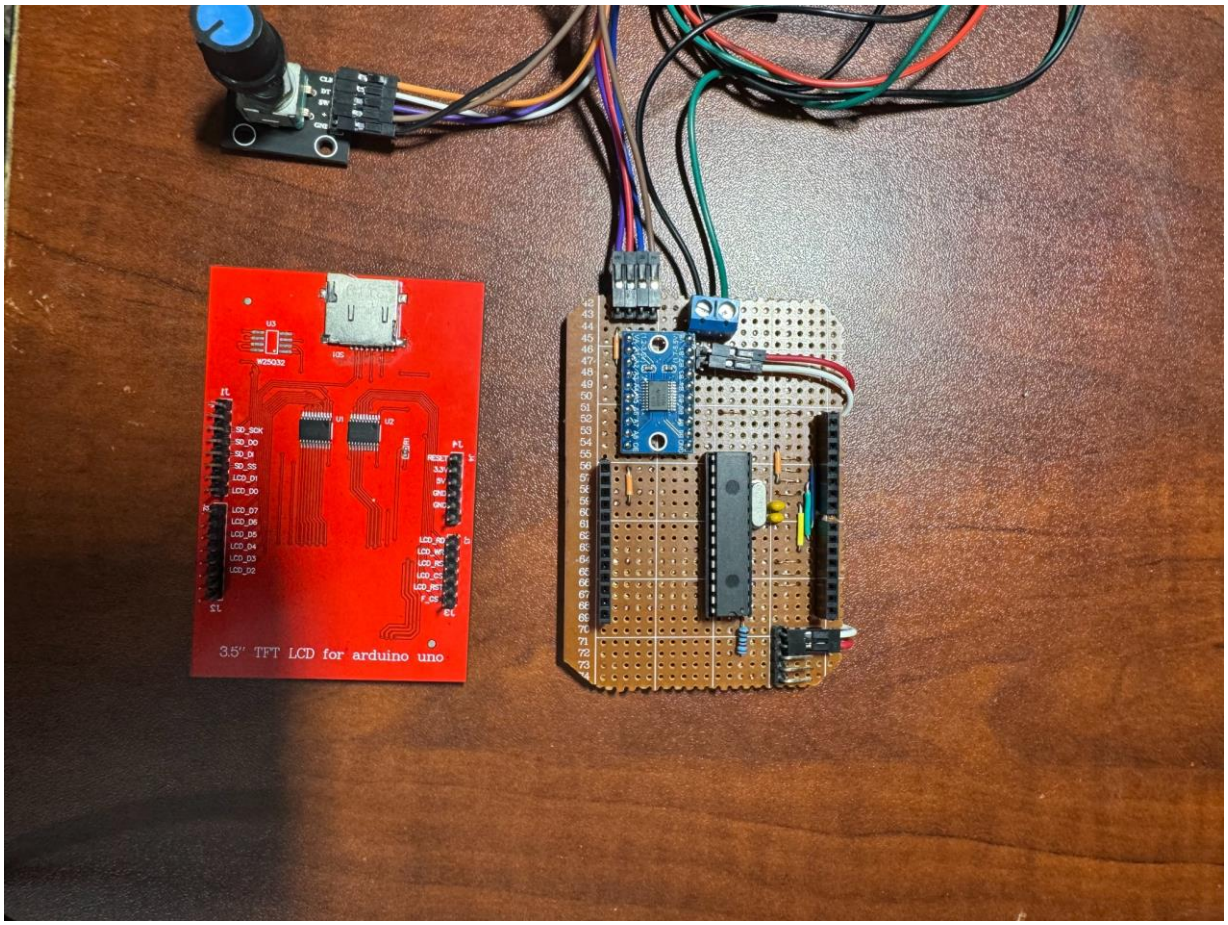


Figure 5. Prototype ATmega328P interface board and rear view of the 3.5-inch TFT module, with rotary encoder and supporting interface circuitry.

The second photograph provides a closer view of the display interface and ATmega328P-based prototype circuitry, illustrating the wiring arrangement used during hardware development. The final PCB implementation is intended to replace these temporary prototype interconnections.